ATOM-SG Exam-Quality Baseline Test v2.0

P6 Mathematics | 40 Questions | 1 Hour | No Calculator | Exam Standard

Name: _____ **Class:** _____ **Date:** _____

Instructions: Show all working. Professional diagrams included. Linguistically complex problems.

Q 1. [H] [WP] Cross-Thread

Mrs. Lee received $240 as a birthday gift from her family. At the department store, she spent 2/5 of this amount on a designer handbag. From the remainder, she then spent 1/3 on a pair of shoes, leaving her with just enough for dinner. If she had $72 left after all her purchases, how much money did she have at first before receiving the gift?

Working:

Answer: _____ **(4 marks)**

Q 2. [H] [WP] Cross-Thread

A rectangular water tank at a factory was filled to 5/6 of its capacity at the start of the day. During the morning shift, workers used 15 litres of water for cleaning, causing the water level to drop to exactly 2/3 of the tank's capacity. Assuming the tank was completely full at the beginning of the week, what is the total capacity of the tank in litres?

Working:

Answer: _____ **(3 marks)**

Q 3. [H] [WP] Cross-Thread

In a sports equipment room, the ratio of red balls to blue balls was 3:5 at the beginning of the term. After 12 red balls were removed for a tournament and not returned, the new ratio of red to blue balls became 1:3. How many red balls were there in the equipment room at the beginning of the term?

Working:

Answer: _____ **(4 marks)**

Q 4. [H] [WP] Cross-Thread

Alan and Ben have been friends since primary school. Currently, their ages are in the ratio 4:7. Six years from now, when Alan enters secondary school, the ratio of their ages will be 2:3. How old is Alan now?

Working:

Answer: _____ **(4 marks)**

Q 5. [H] [WP] Cross-Thread

A stationery shop received a shipment of 240 pens on Monday morning. That day, they sold 2/5 of the pens to a school. On Tuesday, they sold 1/3 of what remained to walk-in customers. On Wednesday, they sold 1/2 of the new remainder to an office. How many pens were left unsold at the end of Wednesday?

Working:

Answer: _____ **(4 marks)**

Q 6. [H] [WP] Cross-Thread

Alice and Betty had $280 in their shared savings jar. One Saturday, Alice spent 1/3 of her personal share on a book, while Betty spent $40 on art supplies. Surprisingly, they discovered they had equal amounts left in their individual portions. How much did Alice originally contribute to the savings jar?

Working:

Answer: _____ **(4 marks)**

Q 7. [H] [WP] Cross-Thread

A baker prepared 360 cupcakes for a weekend market. In the morning session, he sold 1/4 of the cupcakes to early customers. In the afternoon, he sold 2/3 of the remaining cupcakes, which was more than he expected. How many cupcakes did the baker have left to take home at the end of the day?

Working:

Answer: _____ **(3 marks)**

Q 8. [H] [WP] Cross-Thread

Three cousins collect stamps. John has 3/5 as many stamps as his older sister Mary. Mary, in turn, has 5/6 as many stamps as their eldest cousin Tom. If Tom has proudly collected 72 stamps from different countries, how many stamps does John have in his collection?

Working:

Answer: _____ **(3 marks)**

Q 9. [M] [WP] Part-Whole

In Primary 6A, the ratio of boys to girls is 5:3. When the teacher took attendance last Monday, she counted exactly 40 students present in class. How many boys are there in Primary 6A?

Working:

Answer: _____ **(2 marks)**

Q10. [M] [WP] Before-After

A public bus started its route with 60 passengers on board. At the first bus stop, 15 passengers alighted to go to the market. At the second stop, 8 more passengers got off at the school. How many passengers remained on the bus after the second stop?

Working:

Answer: _____ **(2 marks)**

Q11. [H] [WP] Cross-Thread

Ali and Ben had $280 between them in their piggy banks. After Ali kindly gave 1/4 of his savings to Ben to help him buy a toy, they were surprised to find they had exactly the same amount. How much did Ali have in his piggy bank at first?

Working:

Answer: _____ **(4 marks)**

Q12. [H] [WP] Cross-Thread

A water tank was completely full at the start of the day. During the morning, 3/7 of the water was used for cleaning. In the afternoon, 2/5 of the remaining water was used for gardening. At the end of the day, exactly 48 litres were left in the tank. How much water was in the tank at the beginning of the day?

Working:

Answer: _____ **(4 marks)**

Q13. [M] [WP] Part-Whole

There are 80 students registered for the school camp. When the coordinator checked the list, she noticed that 3/8 of the participants are girls. How many boys are going on the camp?

Working:

Answer: _____ **(2 marks)**

Q14. [H] [WP] Cross-Thread

A fruit seller had 240 apples in his stall at the morning market. He sold 3/4 of them to Shop A before noon. Later in the afternoon, he sold 1/2 of the remaining apples to Shop B. How many apples did the seller have left at the end of the day?

Working:

Answer: _____ **(3 marks)**

Q15. [M] [WP] Part-Whole

Jenny and Kelly have been collecting stickers together. In total, they have 84 stickers in their album. Jenny has 16 more stickers than Kelly. How many stickers does Kelly have?

Working:

Answer: _____ **(2 marks)**

Q16. [H] [WP] Cross-Thread

A designer bag's price increased by 20% due to high demand. During a sale two weeks later, the shop offered a 15% discount on the new price. If a customer paid $204 for the bag during the sale, what was the original price before the increase?

Working:

Answer: _____ **(4 marks)**

Q17. [H] [WP] Part-Whole

At a fruit stall, the ratio of apples to oranges to pears is 2:3:5. One morning, the stall owner counted and found that the oranges and pears together total 120 fruits. How many apples are on display?

Working:

Answer: _____ **(3 marks)**

Q18. [H] [WP] Cross-Thread

Mrs. Tan went grocery shopping with some money in her purse. She spent $25 on fresh vegetables. After that, she deposited 1/4 of what remained into her savings. If she had $45 left in her purse for other expenses, how much did she start with?

Working:

Answer: _____ **(3 marks)**

Q19. [M] [WP] Part-Whole

In the school choir, the ratio of boys to girls is 5:3. The conductor noticed there are 24 more boys than girls in the group. How many children are in the choir altogether?

Working:

Answer: _____ **(3 marks)**

Q20. [H] [WP] Cross-Thread

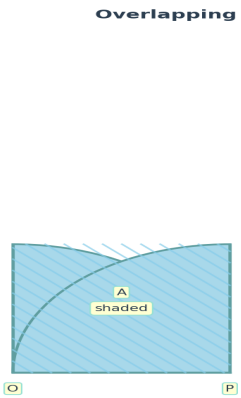
The school library had 450 books on its shelves at the start of the term. During a donation drive, they received 120 new books from parents. Later, they donated 1/5 of their total collection to a sister school. How many books remained in the library?

Working:

Answer: _____ **(3 marks)**

Q21. [H] [G] G5-Composite Overlap

The figure shows two overlapping quarter circles with radius 10 cm. The shaded area OBC is 30 cm². (a) Find the area of figure OABCD. (b) Find the perimeter of figure OABCD.

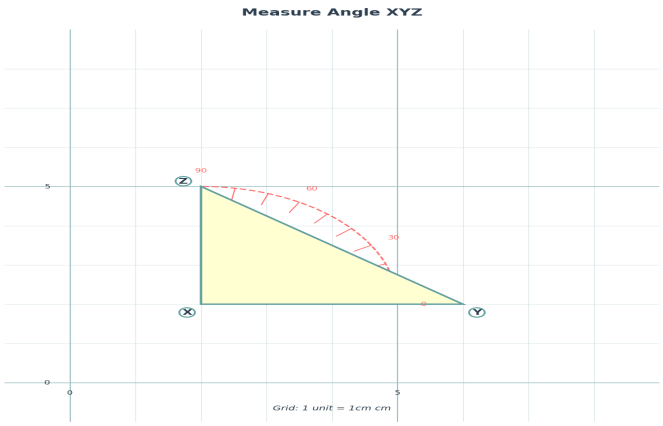


Working:

Answer: _____ **(5 marks)**

Q22. [H] [G] G6-Grid Construction

Triangle ABC is drawn on the grid. (a) Measure angle ACB. (b) Complete trapezium BCDE where BC is parallel to DE, AB = BE, and DE is twice as long as BC.

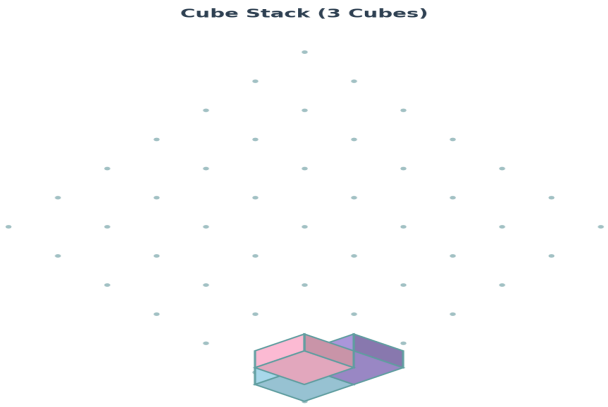


Working:

Answer: _____ **(4 marks)**

Q23. [H] [G] G7-3D Visualization

The solid is made up of 8 cubes. (a) Draw the front view of the solid on the grid. (b) What is the maximum number of cubes that can be added without changing the front and side views?



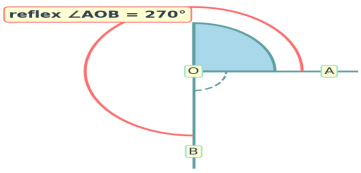
Working:

Answer: _____ **(4 marks)**

Q24. [H] [G] G8-Angle Chasing

In the figure, ABCD is a rhombus and ADEF is a trapezium with AF parallel to DE. Given angle DFE = 21°, angle BCD = 108°, and angle DAB = 33°, find angles x and y.

Reflex Angle at Point O



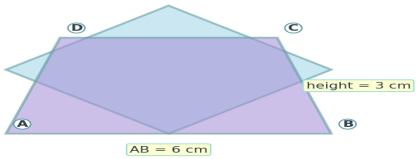
Working:

Answer: _____ **(5 marks)**

Q25. [M] [G] G5-Composite Overlap

Five squares are drawn inside a rectangle. Square X has side 4 cm. Find (a) the side of square Y, (b) the length of the rectangle, (c) how many 2cm cubes can be made from the five squares.

Composite Shape: Rhombus and Trapezium

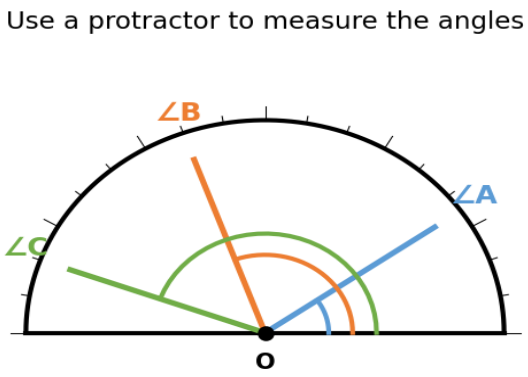


Working:

Answer: _____ **(5 marks)**

Q26. [E] [G] G1-Angle Reasoning

Use a protractor to measure angles A, B, and C in the diagram. Write your answers in degrees.



Working:

Answer: _____ **(3 marks)**

Q27. [M] [G] G8-Angle Chasing

Three angles meet at a point O. Two of the angles measure 120° and 85°. Find the measure of the third angle.

Working:

Answer: _____ **(2 marks)**

Q28. [M] [G] G1-Angle Reasoning

Two straight lines intersect at point O. If one of the angles formed is 40°, find the measures of the other three angles.

Working:

Answer: _____ **(2 marks)**

Q29. [M] [G] G2-Area

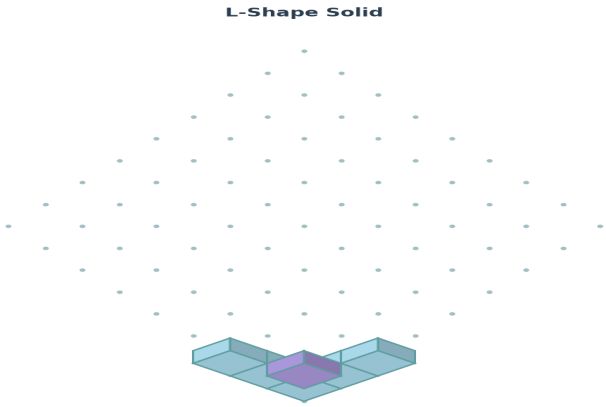
In triangle ABC, angle A measures 50° and angle B measures 70°. Find the measure of angle C.

Working:

Answer: _____ **(2 marks)**

Q30. [M] [G] G2-Composite

The figure shows an L-shaped figure made by combining two rectangles. Find its total area and perimeter.



Working:

Answer: _____ **(3 marks)**

Q31. [M] [G] G2-Area

A rectangular metal sheet measures 12 cm by 8 cm. A smaller rectangular piece measuring 4 cm by 3 cm is cut out from one corner. Find the area of the remaining metal sheet.

Working:

Answer: _____ **(3 marks)**

Q32. [M] [G] G3-Volume

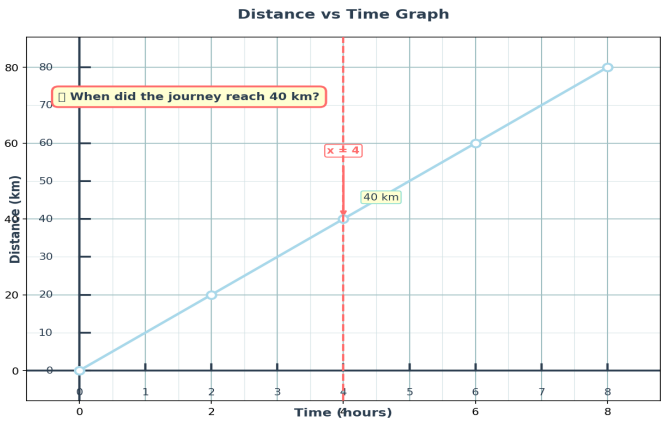
Find the volume of a cuboid with length 6 cm, breadth 4 cm, and height 3 cm.

Working:

Answer: _____ **(2 marks)**

Q33. [H] [DI] DI1-Line Graph

The line graph shows the number of T-shirts left unsold at a shop over 7 days. (a) Which day had the most sales? (b) What percentage of the original stock were sold in the first 5 days? (c) On Day 3, the shop collected $192 after giving a 20% discount. What was the original price per T-shirt before the discount?

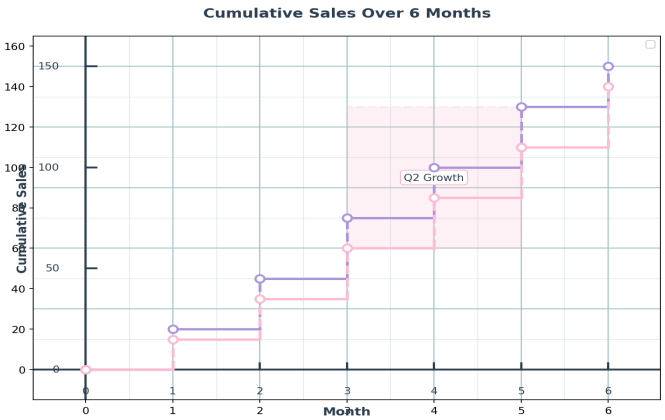


Working:

Answer: _____ **(6 marks)**

Q34. [M] [DI] DI1-Line Graph

Using the line graph showing T-shirt sales, find the total number of T-shirts sold in the first 3 days and calculate the range of daily sales.

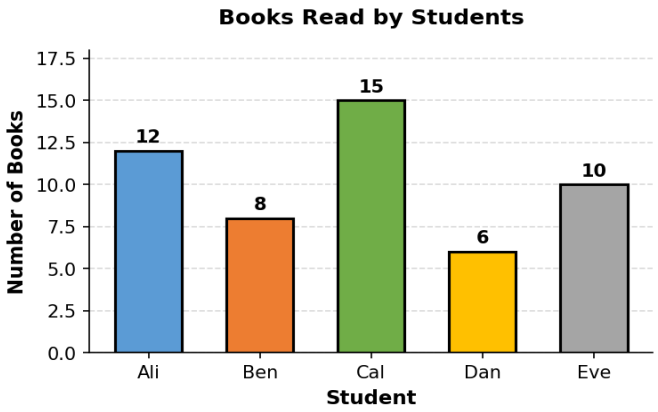


Working:

Answer: _____ **(3 marks)**

Q35. [M] [DI] DI2-Bar Graph

The bar chart shows the number of books read by 5 students during the holidays. (a) Which student read the most books? (b) Calculate the average number of books read. (c) What percentage of the total books were read by Ali and Ben together?

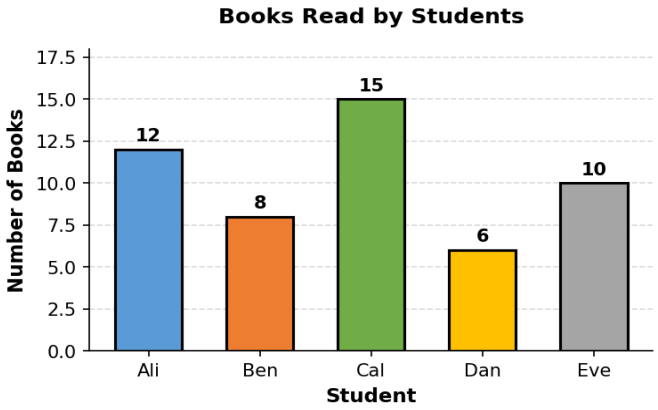


Working:

Answer: _____ **(5 marks)**

Q36. [E] [DI] DI2-Bar Graph

Using the bar chart showing books read by students, find the total number of books read by all 5 students and calculate the range of the data.

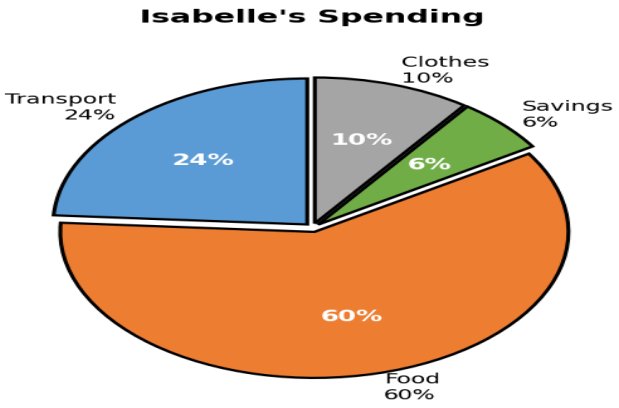


Working:

Answer: _____ **(2 marks)**

Q37. [M] [DI] DI3-Pie Chart

The pie chart shows how Isabelle spent her monthly allowance. She spent 24% on transport, which was exactly 2/5 of what she spent on food. Her savings was 25% of her transport spending. (a) What percentage of her allowance was spent on food? (b) If she saved $90, what was her total monthly allowance?

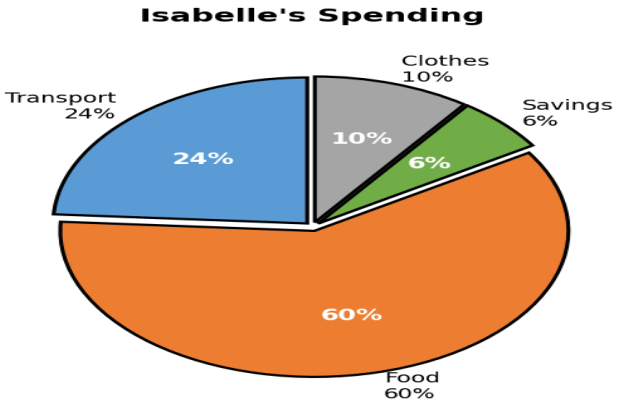


Working:

Answer: _____ **(5 marks)**

Q38. [H] [DI] DI3-Pie Chart

In a pie chart showing fruit distribution, the sector representing apples measures 72° and corresponds to exactly 60 apples. Find the total number of fruits represented in the pie chart.

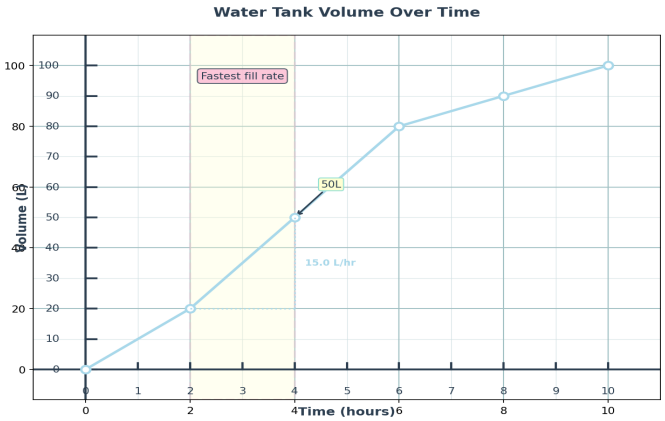


Working:

Answer: _____ **(3 marks)**

Q39. [M] [DI] DI1-Line Graph

Using the temperature data shown in the line graph, estimate what the temperature might have been on Tuesday afternoon and explain your reasoning.

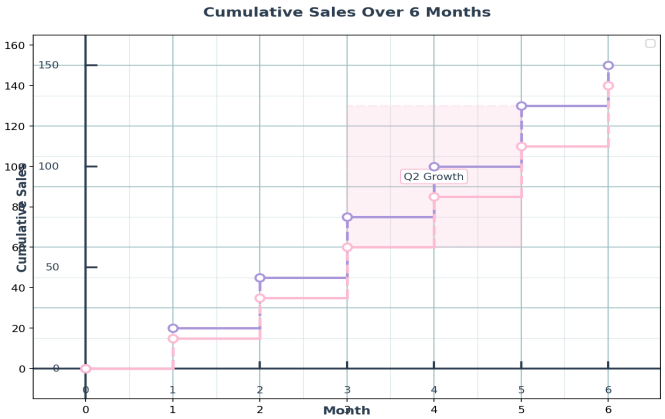


Working:

Answer: _____ **(3 marks)**

Q40. [H] [DI] DI1-Line Graph

Calculate the average temperature for the entire week using the line graph data, and identify which days had temperatures above this average.



Working:

Answer: _____ **(3 marks)**

Answer Key - For Markers Only

Q 1: $480
Q 2: 90 litres
Q 3: 27 red balls
Q 4: 24 years old
Q 5: 48 pens
Q 6: $144
Q 7: 90 cupcakes
Q 8: 36 stamps
Q 9: 25 boys
Q10: 37 passengers
Q11: $160
Q12: 140 litres
Q13: 50 boys
Q14: 30 apples
Q15: 34 stickers
Q16: $200
Q17: 30 apples
Q18: $85
Q19: 96 children
Q20: 456 books
Q21: (a) 127 cm^2 (b) 45.4 cm
Q22: (a) 36° (b) See diagram
Q23: (a) See diagram (b) 8 cubes
Q24: x = 18°, y = 126°
Q25: (a) 2 cm (b) 17.5 cm (c) 38 cubes
Q26: A = 35°, B = 110°, C = 160°
Q27: 155°
Q28: 40°, 140°, 140°
Q29: 60°
Q30: Area = 64 cm^2, Perimeter = 34 cm
Q31: 84 cm^2
Q32: 72 cm^3
Q33: (a) Day 4 (b) 83% (c) $15
Q34: Total: 60, Range: 32
Q35: (a) Cal (b) 10.2 (c) 39.2%
Q36: Total: 51, Range: 9
Q37: (a) 60% (b) $1500
Q38: 300 fruits
Q39: ~30°C with explanation
Q40: Avg: 30.4°C